

Multiple Choice (10 marks)

1. Which of the following decimal numbers has an exact representation in binary notation?

- (a) 0.1
- (b) 0.2
- (c) 0.3
- (d) 0.5

2. Which of the following language features requires that stack-based storage allocation be used rather than static allocation?

- (a) Recursive procedures
- (b) Arbitrary goto's
- (c) Two-dimensional arrays
- (d) Integer-valued functions

3. A 3-way, set-associative cache is

- (a) one in which each main memory word can be stored at any of 3 cache locations
- (b) effective only if 3 or fewer processes are running alternately on the processor
- (c) possible only with write-back
- (d) faster to access than a direct-mapped cache

4. Church's thesis equates the concept of "computable function" with those functions computable by thesis?

- (a) It was first proven by Alan Turing.
- (b) It has not yet been proven, but finding a proof is a subject of active research.
- (c) It can never be proven.
- (d) It is now in doubt because of the advent of parallel computers.

5. Of the following sorting algorithms, which has a running time that is LEAST dependent on the initial ordering of the input?

- (a) Insertion sort
- (b) Quicksort
- (c) Merge sort
- (d) Selection sort

True / False (10 marks)

Answer True or False

6. True or False: In a complete K-ary tree of depth N, the ratio of nonterminal nodes to total nodes is $\frac{K-1}{K}$.
7. True or False: The addition of the two's-complement numbers 10000001 and 10101010 results in 00101011.
8. True or False: Local caching of files in distributed file systems always results in increased network traffic.
9. True or False: The one-time pad is considered a perfectly secure encryption scheme when used correctly.
10. True or False: The access matrix approach, if stored directly, can be large and cumbersome to manage.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to iterate over elements repeating each as many times as its count.
12. Write a python function to find minimum sum of factors of a given number.

End of Paper